CST 3526 Stochastic Process Lecture 5 - 10/22/2026

Lecture 5: Markov Processes I

Lecturer: Ziqiao Wang

In this lecture,[1] we study processes for which the current state contains the information needed to predict the future. Our main focus is on *discrete-time Markov chains*: how to choose a state, specify a model, and calculate multi-step probabilities.

1 Sum Processes: Counts and Random Walks

Let $V_1, V_2, \ldots$ be i.i.d. real random variables with mean μ and variance $\sigma^2 \in (0, \infty)$. Define

$$S_0 = 0, \qquad S_n = \sum_{j=1}^{n} V_j, \qquad n \geq 1.$$

The process V_n records a new increment, while S_n records the accumulated total.

1.1 Dependence comes from shared increments

Immediately,

$$m_S(n) = n\mu, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

For $0 \leq n \leq m$, write $S_m = S_n + (V_{n+1} + \cdots + V_m)$. The new increments are independent of S_n, so

$$C_S(n, m) = \mathrm{Cov}(S_n, S_m) = \mathrm{Var}(S_n) = n\sigma^2.$$

By symmetry, for all nonnegative integers n, m,

$$C_S(n, m) = \sigma^2 \min\{n, m\}, \qquad R_S(n, m) = \sigma^2 \min\{n, m\} + nm\mu^2. \tag{1}$$

The covariance counts the shared increments, each contributing σ^2. In particular, S_n and S_m are dependent for $n, m \geq 1$, since their covariance is positive. For $1 \leq n \leq m$,

$$\rho_S(n, m) = \frac{n\sigma^2}{\sqrt{n\sigma^2\, m\sigma^2}} = \sqrt{\frac{n}{m}}.$$

There is no correlation coefficient involving S_0, whose variance is zero.

Example 1 (Bernoulli counting process)**.** If $V_j = B_j$ are i.i.d. Bernoulli(p), $0 < p < 1$, write the count as $N_n = \sum_{j=1}^{n} B_j$. It has state space $\{0, 1, 2, \ldots\}$ and

$$N_n \sim \mathrm{Binomial}(n, p), \quad m_N(n) = np, \quad C_N(n, m) = p(1-p)\min\{n, m\},$$

$$R_N(n, m) = p(1-p)\min\{n, m\} + p^2 nm.$$

Example 2 (Simple symmetric random walk)**.** If the increments instead take -1 and 1 with equal probability, write $W_n = \sum_{j=1}^{n} V_j$. Then

$$m_W(n) = 0, \qquad C_W(n, m) = R_W(n, m) = \min\{n, m\}, \qquad \mathrm{Var}(W_n) = n.$$

The overall state space is $\mathbb{Z}$, while the reachable states at time n are $\{-n, -n+2, \ldots, n\}$.

[1] Main reading: Leon-Garcia, Sections 11.1–11.3.

2 The Markov Property and the Choice of State

Suppose a system has been observed at times $0, 1, \ldots, n$. In general, predicting its next value requires the joint law of the whole observed history. A Markov model makes a particular simplification: once the present state is known, the earlier states do not change the conditional distribution of the next state.

Definition 2.1 (Discrete-time Markov chain)**.** A process $\{X_n : n = 0, 1, 2, \ldots\}$ with a finite or countably infinite state space S is a *Markov chain* if

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, \ldots, X_{n-1} = i_{n-1}, X_n = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \tag{2}$$

for every n, every $j \in S$, and every history on the left with positive probability.

The equality concerns *conditional distributions*, and does not say that X_{n+1} is independent of X_n: the current state will usually affect the next-state probabilities. Iterating Eq. (2) gives the corresponding statement for any finite block of future states.

Example 3 (A count with independent increments.)**.** Let $B_1, B_2, \ldots$ be independent Bern(p) variables, where $0 < p < 1$, and define

$$N_0 = 0, \qquad N_n = \sum_{r=1}^{n} B_r.$$

Given any feasible history ending at $N_n = i$, the new variable B_{n+1} is still independent of that history. Since $N_{n+1} = i + B_{n+1}$,

$$\mathbb{P}(N_{n+1} = j \mid N_0, \ldots, N_n = i) = \begin{cases} p, & j = i + 1, \\ 1 - p, & j = i, \\ 0, & \text{otherwise.} \end{cases}$$

Thus N_n is Markov.

Example 4. Let $B_0, B_1, \ldots$ be independent fair Bernoulli variables, and observe only

$$Y_n = B_n + B_{n+1}, \qquad n \geq 0.$$

The observed state $Y_1 = 1$ does not tell us whether $(B_1, B_2) = (0, 1)$ or $(1, 0)$. Each possibility has conditional probability $1/2$, so

$$\mathbb{P}(Y_2 = 2 \mid Y_1 = 1) = \mathbb{P}(B_2 = 1, B_3 = 1 \mid Y_1 = 1) = \frac{1}{4}.$$

But $Y_0 = 0$ forces $B_0 = B_1 = 0$. If in addition $Y_1 = 1$, then $B_2 = 1$, and hence

$$\mathbb{P}(Y_2 = 2 \mid Y_1 = 1, Y_0 = 0) = \mathbb{P}(B_3 = 1) = \frac{1}{2}.$$

Both conditioning events have positive probability. The earlier observation changes the prediction, so Y_n is not Markov.

3 Transition Probabilities and the Law of a Chain

Definition 3.1 (Time-homogeneous transitions)**.** A Markov chain is *time-homogeneous* if its one-step transition probabilities do not depend on the time index:

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i), \qquad i, j \in S. \tag{3}$$

The Markov property removes dependence on the *earlier history*; time homogeneity removes dependence on the *clock time*. They are different assumptions. For example, take independent Bernoulli variables with success probability $1/4$ at even times and $3/4$ at odd times. They form a Markov chain, but the next-state law alternates with time, so the chain is not time-homogeneous. Unless stated otherwise, the chains below are time-homogeneous.

3.1 The transition matrix

For $S = \{0, 1, \ldots, M-1\}$, arrange the numbers p_{ij} into

$$\mathbf{P} = [p_{ij}]_{i,j \in S}, \qquad p_{ij} \geq 0, \qquad \sum_{j \in S} p_{ij} = 1.$$

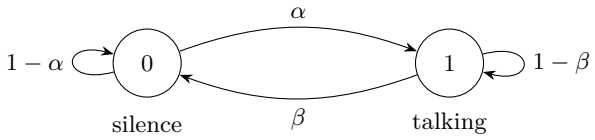
The row is the current state; the column is the next state. Each row is a probability distribution. A directed edge $i \to j$ in the state diagram is drawn when $p_{ij} > 0$ and is labeled by that probability. Self-loops are transitions too. The same row-sum rule applies to countably infinite state spaces.

Example 5 (A two-state speech model.)**.** Let $X_n = 0$ mean silence and $X_n = 1$ mean talking during time slot n. Suppose the next-slot law depends only on the present mode. Write

$$\alpha = \mathbb{P}(X_{n+1} = 1 \mid X_n = 0), \qquad \beta = \mathbb{P}(X_{n+1} = 0 \mid X_n = 1).$$

For $0 < \alpha, \beta < 1$, the transition matrix in state order $(0, 1)$ is

$$\mathbf{P} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}. \tag{4}$$



We will use $\alpha = 1/4$, $\beta = 1/2$ for numerical calculations:

$$\mathbf{P} = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}. \tag{5}$$

The conditional speech probability is $1/4$ after silence and $1/2$ after talking. The model is therefore Markov.

3.2 Initial probabilities and path probabilities

Let $p_i(0) = \mathbb{P}(X_0 = i)$. The initial distribution is an essential part of the specification: $\mathbf{P}$ tells us how the system moves, not where it starts. By the chain rule and the Markov property,

$$\begin{aligned} \mathbb{P}(X_0 = i_0, X_1 = i_1, \ldots, X_n = i_n) &= p_{i_0}(0) \prod_{r=1}^{n} \mathbb{P}(X_r = i_r \mid X_0 = i_0, \ldots, X_{r-1} = i_{r-1}) \\ &= p_{i_0}(0)\, p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n}. \end{aligned} \tag{6}$$

The derivation applies directly to positive-probability histories; any impossible path has probability zero. Thus an initial distribution and a transition matrix determine all finite-dimensional distributions of the chain.

For Eq. (5), suppose $X_0 = 0$ with probability one. The probability of the particular path $(0, 1, 1, 0)$ is

$$\mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 1, X_3 = 0) = p_{01} p_{11} p_{10} = \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}.$$

Note that this is not the probability of simply being in state 0 at time 3: many other paths have that same endpoint. To obtain an endpoint probability, we must add the probabilities of all admissible intermediate paths.

4 Multi-Step Transitions and Chapman–Kolmogorov

Define the n-step transition probability by

$$p_{ij}(n) = \mathbb{P}(X_{m+n} = j \mid X_m = i), \qquad p_{ij}(0) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Time homogeneity makes this independent of m. In particular, $p_{ij}(1) = p_{ij}$. The argument n is the *number of steps*, not the time at which a one-step transition occurs.

Definition 4.1 (Chapman–Kolmogorov equations)**.** For nonnegative integers m, n,

$$p_{ij}(m+n) = \sum_{k \in S} p_{ik}(m) p_{kj}(n). \tag{7}$$

Proof. Condition on the intermediate state $X_m = k$, starting from $X_0 = i$. The law of total probability gives

$$\begin{aligned} p_{ij}(m+n) &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j \mid X_m = k, X_0 = i) \mathbb{P}(X_m = k \mid X_0 = i) \\ &= \sum_{k \in S} p_{kj}(n) p_{ik}(m). \end{aligned}$$

The second equality uses the multi-step form of the Markov property and time homogeneity. Terms with zero intermediate probability contribute zero. □

If $\mathbf{P}(n) = [p_{ij}(n)]$, then Eq. (7) is matrix multiplication:

$$\mathbf{P}(m+n) = \mathbf{P}(m)\mathbf{P}(n), \qquad \mathbf{P}(0) = \mathbf{I}, \qquad \mathbf{P}(n) = \mathbf{P}^n.$$

An entry of $\mathbf{P}^n$ sums products of transition probabilities along paths of length n.

Recall the speech model after two steps, for Eq. (5),

$$\mathbf{P}^2 = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix} = \begin{pmatrix} 11/16 & 5/16 \\ 5/8 & 3/8 \end{pmatrix}.$$

For example,

$$p_{01}(2) = p_{00}p_{01} + p_{01}p_{11} = \frac{3}{4}\frac{1}{4} + \frac{1}{4}\frac{1}{2} = \frac{5}{16}.$$

The two terms correspond to the disjoint paths $0 \to 0 \to 1$ and $0 \to 1 \to 1$. Notice that $p_{01}(2) \neq (p_{01})^2$.

4.1 Propagating the distribution

Use a *row vector* for state probabilities:

$$\mathbf{p}(n) = (p_0(n), p_1(n), \ldots), \qquad p_j(n) = \mathbb{P}(X_n = j).$$

Conditioning on X_n gives

$$p_j(n+1) = \sum_i p_i(n) p_{ij}, \qquad \mathbf{p}(n+1) = \mathbf{p}(n)\mathbf{P}, \qquad \mathbf{p}(n) = \mathbf{p}(0)\mathbf{P}^n.$$

For the speech model starting in silence, $\mathbf{p}(0) = (1, 0)$ and

$$\mathbf{p}(1) = (3/4, 1/4), \qquad \mathbf{p}(2) = (11/16, 5/16).$$

The two-time law is obtained in the same way:

$$\mathbb{P}(X_m = i, X_{m+n} = j) = p_i(m) p_{ij}(n). \tag{8}$$

Thus transition probabilities connect the model to the finite-dimensional distributions introduced in Lecture 4.

5 A Modeling Example: Exchanging Balls Between Two Urns

There are two white balls and two black balls in total. Each urn contains two balls. At every step, choose one ball uniformly from each urn, independently of the other choice, and exchange the selected balls. Choices at a new step use fresh randomness.

Let X_n be the number of white balls in the first urn after n exchanges. Then $S = \{0, 1, 2\}$. When $X_n = i$, the urn compositions are

	white	black
urn 1	i	$2 - i$
urn 2	$2 - i$	i

Because these counts determine the distribution of the next exchange, the count X_n is a sufficient state; we do not need to remember the individual balls or earlier exchanges.

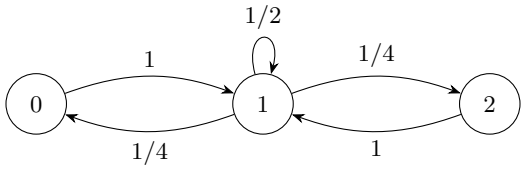
The count increases by one if a black ball leaves urn 1 and a white ball enters it. It decreases by one for the reverse exchange. Therefore, for feasible destinations,

$$p_{i,i+1} = \left(\frac{2-i}{2}\right)^2, \qquad p_{i,i-1} = \left(\frac{i}{2}\right)^2, \qquad p_{ii} = 2\frac{i}{2}\frac{2-i}{2}.$$

At $i = 0$ and $i = 2$, the formulas for an outward move are zero. In state order $(0, 1, 2)$,

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1 & 0 \end{pmatrix}. \tag{9}$$

Every row sums to one. From state 0 or state 2, the next count is certainly 1. From state 1, an exchange of balls of the same color leaves the count unchanged, with probability $1/2$.



6 Communication, Returns, and Periodicity

We say that j is *accessible* from i, and write $i \to j$, if $p_{ij}(n) > 0$ for some $n \geq 0$. States i and j *communicate*, written $i \leftrightarrow j$, if each is accessible from the other. Communication partitions S into communicating classes. A chain is *irreducible* if all its states belong to one communicating class.

A class is *closed* if no transition leaves it. A state i is *absorbing* if $p_{ii} = 1$; its singleton class is closed.

For the urn chain Eq. (9), states 0 and 2 communicate through state 1. All three states form one closed class, so the chain is irreducible, despite the zero entries $p_{02} = p_{20} = 0$.

6.1 Recurrence and Transience

Assume $X_0 = i$. The *first return time*

$$T_i^+ = \inf\{n \geq 1 : X_n = i\}, \qquad \inf \varnothing = \infty,$$

is the first strictly positive time at which the chain is in state i again. Here the infimum is simply the smallest qualifying integer. For the path $i \to j \to k \to i$, where $j, k \neq i$, we have $T_i^+ = 3$. If the next step is a self-loop $i \to i$, then $T_i^+ = 1$. If it never returns, $T_i^+ = \infty$.

There are two questions: *Will a return occur?* and *How long is the average wait?* State i is *recurrent* if

$$\mathbb{P}(T_i^+ < \infty \mid X_0 = i) = 1.$$

It is *transient* if this probability is less than one: there is a positive probability of never returning. Transient does not mean that a return is impossible. A recurrent state is *positive recurrent* if, in addition,

$$\mathbb{E}[T_i^+ \mid X_0 = i] < \infty.$$

Returning with probability one does not by itself guarantee a finite mean waiting time. It also does not mean returning before some fixed deadline with probability one.

In a *finite* chain, states in closed communicating classes are positive recurrent; states outside those classes are transient. In particular, **all states of a finite irreducible chain are positive recurrent**. Note that finiteness matters: on an infinite state space, irreducibility alone does not guarantee recurrence or a finite mean return time.

Example 6 (An absorbing chain.)**.** Consider the constructed chain, in state order $(0, 1, 2)$,

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 0 & 1 \end{pmatrix}. \tag{10}$$

Its three communicating classes are $\{0\}$, $\{1\}$, and $\{2\}$. States 0 and 2 are absorbing. Starting from 1, there are two possibilities at the first step:

first step	probability	first return time to 1
$1 \to 1$	$1/2$	$T_1^+ = 1$
$1 \to 0$ or $1 \to 2$	$1/2$	$T_1^+ = \infty$

Once the chain leaves 1, it is trapped at 0 or 2 and cannot return. Thus

$$\mathbb{P}(T_1^+ < \infty \mid X_0 = 1) = \frac{1}{2} < 1,$$

so state 1 is transient, even though a return is possible. By contrast, starting at either absorbing state $i \in \{0, 2\}$ gives $T_i^+ = 1$ with probability one, so both absorbing states are positive recurrent.

6.2 Periodicity

Recurrence asks whether the chain will return; periodicity asks *at which step numbers it can be back*. Recall that

$$p_{ii}(n) = \mathbb{P}(X_n = i \mid X_0 = i).$$

This probability allows earlier visits to i. It is different from $\mathbb{P}(T_i^+ = n \mid X_0 = i)$, which requires the *first* return to occur at step n.

For a state that can return, define its *period* by

$$d(i) = \gcd\{n \geq 1 : p_{ii}(n) > 0\}.$$

Here gcd means the greatest common divisor. Thus we list all possible positive return step numbers and take their greatest common divisor. **The period is not the shortest return time.**

Example 7 (An alternating chain)**.** The transition matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{11}$$

forces the path $0 \to 1 \to 0 \to 1 \to 0 \to \cdots$ when $X_0 = 0$. Hence

$$p_{00}(2n) = 1, \qquad p_{00}(2n+1) = 0, \qquad n \geq 0.$$

The possible positive return step numbers are $2, 4, 6, \ldots$, so

$$d(0) = \gcd(2, 4, 6, \ldots) = 2.$$

State 1 also has period two.

A state is *aperiodic* if $d(i) = 1$. This does not require a return in one step. For example, if both paths $i \to j \to i$ and $i \to j \to k \to i$ have positive probability, returns are possible in two and three steps. Since $\gcd(2, 3) = 1$, the period is one, even if there is no self-loop. Period $d > 1$ means that every possible return step number is a multiple of d; it does not mean that a return occurs at every such multiple.

Self-loop. If $p_{ii} > 0$, then 1 belongs to the set of possible return step numbers. A set containing 1 has greatest common divisor 1, so $d(i) = 1$. **A self-loop is therefore *sufficient*, but not necessary, for aperiodicity.**

Communicating states have the same period. Consequently, one self-loop makes an *irreducible* chain aperiodic. For the urn chain Eq. (9), all three states communicate and $p_{11} = 1/2 > 0$, so all three have period one. The alternating chain instead has period two; this will obstruct convergence from a fixed initial state.

Justification that communicating states have equal periods: If i reaches j in r steps and j reaches i in s steps, then both $r + s$ and $r + n + s$ are return lengths for i whenever n is a return length for j. Hence $d(i)$ divides their difference n. Thus $d(i)$ divides $d(j)$; interchanging i and j gives equality.